

Major-Project Synopsis

IoT-Driven Cold Chain System for Real-Time Monitoring and Predictive Failure Prevention		
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Introduction:

Cold chain logistics are critical for preserving temperature-sensitive products like vaccines, biologics, and perishables. A break in the cold chain due to equipment failure, power outage, or human error can lead to product spoilage, financial loss, and public health risks. Traditional monitoring systems are manual, lack real-time alerts, and cannot predict failures before they occur.

This project proposes an IoT-Driven Cold Chain System using ESP32, DHT22, GSM, and cloud integration via Blynk to enable real-time temperature/humidity monitoring, instant SMS alerts, and door status tracking. The system also incorporates a Peltier-based cooling module with automated control. By logging data to the cloud, the system enables predictive analytics to flag potential failures, ensuring compliance with WHO-recommended 2–8°C storage norms for vaccines.

Literature Review:

[1] The study titled “Enhancing Food Safety in the Cold Chain Through Internet of Things and Artificial Intelligence” by Wagner Augusto Müller, Sandro Binsfeld Ferreira, and Suse Botelho da Silva in the Journal of Food Science (2026) analyzes how different storage conditions impact food quality and shelf life. It emphasizes the role of IoT and AI in monitoring and evaluating changes in food quality under varying temperature conditions. However, the work highlights limitations such as high cost, complexity, and limited real-world deployment, showing the need for integrated closed-loop systems.

[2] The paper “Development of disposable paper-based smart visual labels with thermochromic materials for cold chain temperature (–18 °C) monitoring” by Qiwen Chen, Yulun Chen, Qingrun Liu, and Shaofeng Yuan in ScienceDirect (2025) proposes a low-cost solution using thermochromic labels that permanently alter QR code readability when temperature thresholds are exceeded. This provides a simple visual indication of cold chain failure. The limitation is that the system is single-use, prone to damage, and lacks real-time monitoring or alert capabilities.

[3] The research “Predicting ‘broken chain’ risks in food and pharmaceutical cold chains based on multi-modal data fusion and machine learning” by Gang Ping in the Journal of Hunan University Natural Sciences (2025) introduces advanced predictive models using IoT data, GPS, weather inputs, and machine learning (LSTM & Transformer) to forecast temperature deviations and generate early alerts. While effective in theory, the approach depends heavily

on high-quality data and faces challenges in real-time deployment and maintaining model accuracy.

[4] The paper “Development of an internet of things based smart cold storage with inventory monitoring system” by Suganya Angappan, Aarthi Nataraj, Loganathan Navaneetha krishnan, and Anbaradu Palanisamy in the International Journal of Electrical and Computer Engineering (2024) discusses IoT-based smart cold storage systems integrated with inventory monitoring. The work highlights the potential of combining environmental monitoring with stock management to improve efficiency. However, limitations include incomplete integration details and the need for further validation in practical deployments.

[5] The paper “IoT-Based Integrated Sensing and Logging Solution for Cold Chain Management” published in IEEE Xplore (2025) presents a comprehensive framework for monitoring temperature-sensitive goods using IoT sensors and cloud-based logging. The study highlights how integrated sensing devices can continuously capture environmental parameters such as temperature and humidity, while cloud storage ensures traceability and compliance with safety standards. The system improves visibility across the supply chain but faces challenges related to sensor calibration, data synchronization, and infrastructure costs.

[6] The research “Cold Chain Monitoring with IoT Sensors: A Data-Driven Approach to Reducing Spoilage in Temperature-Sensitive Supply Chains” by Maurice Efiom Eyo, Too hukwu Rufus Okwara, and colleagues in the European Journal of Logistics, Purchasing and Supply Chain Management (2026) investigates IoT-enabled monitoring for perishable goods. The authors propose a conceptual framework integrating IoT sensors, cloud analytics, and decision support systems to mitigate spoilage risks. Using empirical data from food and vaccine logistics, the study demonstrates improved responsiveness and traceability. Limitations include data integration challenges and infrastructure constraints, but the work provides actionable insights for building resilient cold chain systems.

[7] The article “Revolutionizing Cold Chain Logistics: Leveraging IoT and AI for Enhanced Food Safety and Waste Reduction” by Munikumar Pindi in the World Journal of Advanced Research and Reviews (2025) explores the combined use of IoT sensors, AI analytics, and blockchain for cold chain management. It emphasizes predictive quality management, anomaly detection, and route optimization to reduce food waste and improve safety. The integration of blockchain ensures transparency and traceability, while AI transforms raw sensor data into actionable intelligence. The study concludes that these technologies shift cold chain systems from reactive monitoring to proactive prevention, though adoption challenges include cost, technical expertise, and regulatory alignment

Problem Statement:

- Existing cold chain systems mainly focus on monitoring, not prevention.
- Delayed response leads to temperature rise before action is taken
- Lack of humidity and airflow control causes spoilage of fresh produce
- Dependence on manual intervention and internet reduces reliability

Objectives:

- To develop a smart cold chain system for safe storage of vaccines and fresh produce
- To continuously monitor temperature, humidity, and airflow
- To implement automatic control and failure prevention
- To provide real-time alerts and IoT-based monitoring

Challenges:

- Ensuring sensor accuracy in harsh environments.
- Maintaining network reliability (Wi-Fi/GSM).
- Designing energy-efficient cooling control.
- Handling large volumes of data for predictive analytics.
- Providing user-friendly dashboards for both supervisors and operators.

Motivation:

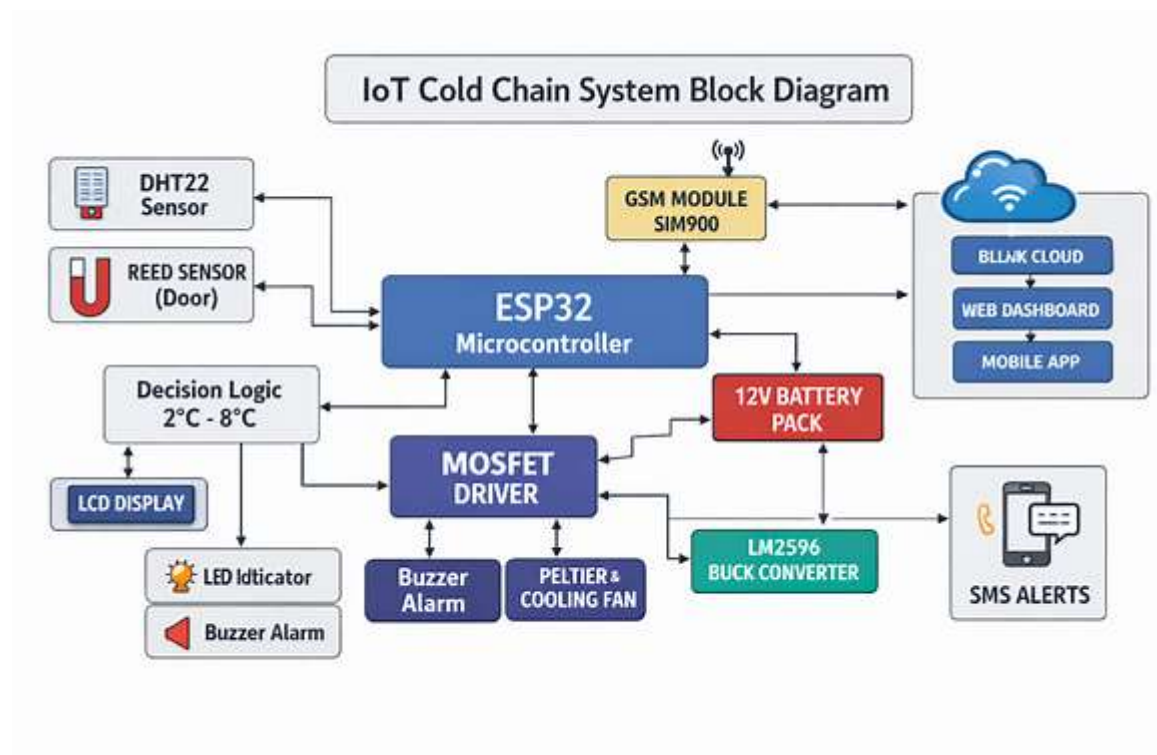
- Rising demand for vaccine distribution and perishable goods logistics.
- Need for cost-effective, scalable, and reliable monitoring solutions.
- Desire to reduce human error and manual dependency.
- Opportunity to integrate predictive analytics for proactive risk management.

Methodology

- Sensor Integration: Connect DHT22, door sensor, relay, buzzer, GSM, and LCD to ESP32.
- Data Acquisition: Read temperature, humidity, and door status at fixed intervals.
- Decision Logic: Compare sensor values against thresholds ($2-8^{\circ}\text{C}$, $\leq 90\% \text{ RH}$).
- Action Execution: Trigger relay, buzzer, and SMS alerts when abnormal conditions occur.

- Cloud Communication: Publish data via MQTT to Blynk Cloud.
- Dashboard Visualization: Display gauges, graphs, and alerts on web and mobile dashboards.
- Predictive Analytics: Use trend analysis to forecast potential failures.
- Testing & Validation: Simulate abnormal conditions and verify system response.

Block Diagram:



Possible Outcomes:

- Accurate real-time monitoring of cold chain parameters.
- Reduced spoilage through predictive alerts and corrective actions.
- Improved reliability with multi-channel notifications.
- Scalable solution for logistics companies managing multiple assets.
- Compliance support for pharmaceutical and food industries.

References:

- [1] Wagner Augusto Müller, Sandro Binsfeld Ferreira, Suse Botelho da Silva, “Enhancing Food Safety in the Cold Chain Through Internet of Things and Artificial Intelligence,” *Journal of Food Science*, 2026.
- [2] Qiwen Chen, Yulun Chen, Qingrun Liu, Shaofeng Yuan, “Development of Disposable Paper-Based Smart Visual Labels with Thermochromic Materials for Cold Chain Temperature (–18 °C) Monitoring,” *ScienceDirect*, 2025.
- [3] Gang Ping, “Predicting ‘Broken Chain’ Risks in Food and Pharmaceutical Cold Chains Based on Multi-Modal Data Fusion and Machine Learning,” *Journal of Hunan University Natural Sciences*, 2025.
- [4] Suganya Angappan, Aarthi Nataraj, Loganathan Navaneetha Krishnanz, Anbaradu Palanisamy, “Development of an Internet of Things Based Smart Cold Storage with Inventory Monitoring System,” *International Journal of Electrical and Computer Engineering (IJECE)*, 2024.
- [5] IEEE Xplore, “IoT-Based Integrated Sensing and Logging Solution for Cold Chain Management,” 2025.
- [6] Maurice Efiom Eyo, Toochukwu Rufus Okwara, et al., “Cold Chain Monitoring with IoT Sensors: A Data-Driven Approach to Reducing Spoilage in Temperature-Sensitive Supply Chains,” *European Journal of Logistics, Purchasing and Supply Chain Management*, 2026.
- [7] Munikumar Pindi, “Revolutionizing Cold Chain Logistics: Leveraging IoT and AI for Enhanced Food Safety and Waste Reduction,” *World Journal of Advanced Research and Reviews*, 2025.

Acceptance/Rejection by the Panel:

Comments:

Accepted/ Rejected